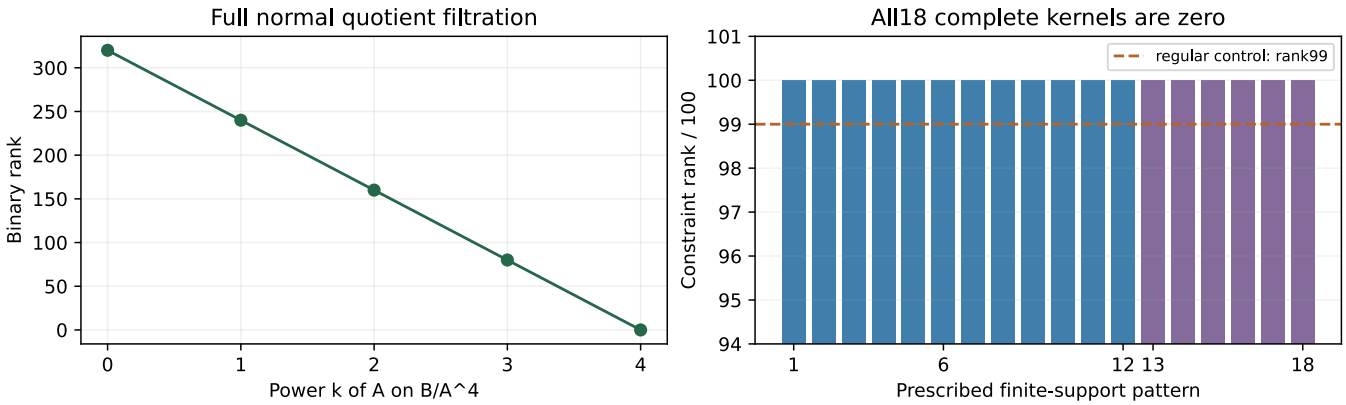
R59: complete normal jets and finite-support exclusions

All18 prescribed F2-stable ordinary O/T fiber patterns have zero solution spaces on the actual complete100-section source system. Explicit left inverses certify every exclusion and persist after extension to F32. A separate exact certificate invocation checks every matrix product using row-dot-column parity rather than the producer's column-XOR method.



Exact recorded counts from cohort59-01, replayed by checkcohort59-01. Patterns1–12 are ordered degree20 component pairs;13–18 are degree40 complementary partitions. The dashed rank99 control has a known nonzero kernel. No statistical or performance comparison is represented.

Complete source-bound quotient

The actual normal algebra B is free of rank32 over F2[z]. Its commuting annihilator satisfies A C=(z^5+z^2+1)I. Hence f^4 B is contained in A^4 B. The presentation modulo f^4 has640 binary coordinates; its complete cokernel is320-dimensional. Explicit products certify

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E M4=0, E G=I320, M4 H+G E=I640
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Four source-polynomial-labelled CRT blocks each have dimension80. Their complete Jordan models have length4. Successive A ranks are320,240,160,80,0. The source operator nu/b and old reduced model agree on all160 retained columns. All100 R58 numerator functions reconstruct in the full normal source field, with3,200 coefficient denominator-unit checks.

Complete prescribed cohort

For x=f/A squared, impose numerator order at least4 at T and at least2 outside O union T. All12 ordered one-component pairs and6 complementary partitions are retained. Every constraint matrix C has rank100 and an exact H C=I100. Thus all18 principal-function spaces vanish over F2 and F32. No31-scalar Artin–Schreier branch remains within this finite cohort. The regular-function positive control has rank99 and contains the actual A squared. A changed cokernel projection is rejected by the checker.

The interpretation as normal fourth-order local jets uses the retained proper-source and reduced-support/DVR premises from R55/R56. It does not prove saturation of the separate incidence or projective ideal. Nonstable F32 supports and other polar divisors remain separate construction spaces.

Failures, formal implications and the next construction

Attempt	Workflow	Outcome
checkcohort59-01	Rust certificate checker	accepted
cohort59-01	Rust producer	accepted
data59-01	Singular	rejected
data59-02	Singular	accepted
jets-lean59-01	Lean	accepted

Five workflows:4 accepted,1 rejected. data59-01 omitted linalg.lib, so unitmat was unavailable. Cascading errors and a late PASS are retained and excluded. data59-02 repairs that import before its fresh invocation. All31 executed-source Git pins match their historical committed bytes.

Five audited generic Lean implications

Jets59.fourth_annihilator transfers the commuting annihilator to fourth powers. fourth_idempotent and fourth_orthogonal justify the thickened CRT formulas under native nilpotence premises. fourth_nilpotent_model transfers the invertible length4 model to A. left_inverse_basechange proves that each native H C=I remains an identity after coefficient extension, including F32.

Lean4.34.1 with mathlib d13f23b723b8a846827a245b89c10fc7d3f11612, -j1 -M4096, separate1200second compilation/audit caps. Both exit0. Five authored declarations are audited; zero failing, zero trusted native_decide. Actual arrays, function-field bindings and source geometry remain explicit native premises. See PROOF_MAP.md and runs/jets-lean59-01/audit.log.

Next: infinity-supported ordinary maps

The four actual degree2 infinity points have original pole caps56,8,14,2. Balanced signed ramification permits404 nonzero patterns. It forces both x and t into the tighter common bound24,8,14,2, of degree96. First construct that complete mixed space and its actual involution. Test BOTH trace parities for t, with a genuine mixed-space particular function when trace1 is feasible. The finite-pole lambda/A is not a valid substitute here.

For every principal x pattern, test all31 F32 scalars, both ordinary signs and both feasible parities with one exact AS operator. Final candidates require whole-source identities, all5 coefficient conjugates, descent and intended order11 action. Degree22 is possible in this cohort and must not be removed by the earlier finite-support degree-unit filter.

Preserve wider F32 supports, direct500-bit relations, L(3P), integral level77 factor geometry/action, proper polarization, separate saturation, node187 coupled divisors and arbitrary reverse classes including460/367. All original native caps and existing managed owners persist. No accepted producer is repeated for publication. NEXT.md records the full next scope.

Sources and reproducibility

CM.md, PROTOCOL.md, RUST-PROTOCOL.md, LEAN-PROTOCOL.md, results.json, export.json and SESSION.md bind source functions, attempted workflows, controls and exact acceptance. Donors retain R52 normal tables, R55 annihilator/CRT, R58 complete spaces and source maps. Singular official source documentation was checked for helper/library behavior. No novelty, performance or independent scientific-review claim is made.